Original Article

Preferences Including Graft Choice, Lateral Augmentation, and Rehabilitation After Anterior Cruciate Ligament Reconstruction Among National Football League Team Orthopaedic Surgeons

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Purpose: To assess current treatment preferences among National Football League (NFL) team orthopaedic knee surgeons for the management of anterior cruciate ligament (ACL) injuries. **Methods:** NFL team physicians who are sports medicine knee surgeons were identified by using public databases and emailed a multiple-choice poll evaluating various preferences involving treating ACL injuries in various situations. **Results:** Ninety-seven NFL team knee surgeons were identified, with 51 responding (53%). Respondent surgeons had 19.6 years of experience performing ACL surgery with 103.9 ACL reconstructions performed annually. Patella tendon was the favored graft in a 20-year-old running back, as well as the 25-year-old and 35-year-old recreational athlete. Quadriceps tendon became the second most favored graft in all situations. In total, 4% of surgeons said they would perform a primary repair of an ACL in the setting of a proximal tear in a 20-year-old running back, 96% said they would consider adding a lateral augmentation procedure in the primary ACLR setting but only if certain findings were obtained, 25% recommended outpatient physical therapy to supplement working with the team rehabilitation staff, whereas 76% did not recommend a brace for return to play. **Conclusions:** Patella tendon autograft is the favored graft in ACL reconstruction among professional and recreational athletes by NFL team orthopaedic knee surgeons. Quadriceps tendon autograft is the second most preferred graft. Lateral augmentation procedures would be considered in a primary ACLR but only if certain findings are present. Most respondents do not recommend supplementing rehab with outpatient PT or using a brace for return to play. **Clinical Relevance:** NFL team surgeons consider numerous factors when treating ACL injuries in professional athletes, including return to function, speed of recovery, and risk of reinjury, all of which directly impact the athlete's ability to sustain a livelihood during a relatively short professional career. Insights gained from these surgical practices may inform and enhance treatment strategies across a broader patient population.

Anterior cruciate ligament (ACL) injury in a competitive, pivoting athlete can have substantial consequences on ability to return to sport at the same level.[1] For professional athletes, this injury can have significant influence on a career, including monetary earning potential.[2] The National Football League (NFL) is one of the most popular entertainment products in North America, and thus its players are more high profile than other athletes.[3] The secondary pressures on providers to provide an optimal outcome and in an efficient amount of time is present but difficult to quantify. The 2014 investigation on preferences of treating ACL injuries by NFL and National Collegiate Athletic Association (NCAA) was a popular paper that helped create a reference point for many providers when explaining ACL surgery choices to patients as well as other providers. The poll itself was presented to the physicians in 2013, with the paper published in 2014.[4]

Our hypothesis is that there would have been notable changes to preferences, including increased

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quadriceps tendon use, decreased hamstring use, and decreased transtibial reaming. The purpose of this study was to assess current treatment preferences among NFL team orthopaedic knee surgeons for the management of ACL injuries. We hypothesized that patella tendon and quadriceps tendon autografts would increase in preference whereas there would be a decrease in preference for hamstring autograft and that nontranstibial femoral tunnel reaming would increase in preference.

Methods

The heads of the NFL Physicians Society were notified of the investigation. The study idea was proposed to the institutional review board, who stated that no formal approval was required. The team surgeons were determined by Internet web searches and then individual profiles investigated to see that they performed sports medicine knee surgery. Inclusion criteria were NFL team physicians who performed knee surgery. Exclusion criteria were NFL team physicians who did not perform knee surgery. Notably, NCAA physicians were excluded from the methods, which makes the investigation different from the 2014 investigation. This was in part due to wanting to separate out NFL team knee surgeons as well as availability of NCAA team physicians. In May 2023, the identified NFL team physicians were emailed with a link to the survey. The authors created an 11-question survey created on the website Google (www.google.com/forms/about). This was designed to be similar to the survey published in 2014.[4] This was sent to surgeons 4 times over May 2023. The survey was designed so that the individual could not complete the survey without completing all questions (Fig 1). All responses were received in May 2023. The responses were kept confidential, and the data were later organized through a spreadsheet. Responses were broken down by individual questions and answers with no advanced statistical analysis. Questions that were present on the 2014 paper[4] were then compared directly.

Results

The survey was sent to the e-mail addresses of 97 NFL team orthopaedic surgeons; 51 team orthopaedic surgeons responded (53%). The average years of experience performing ACL surgery was 19.6 ± 11.4 years (range 4-40 years). Surgeons said that, in 2022, they performed approximately 103.9 ± 46.6 (range 5-250, median 100) ACL reconstructions. For question 3, which asked preferred graft choice for a 20-year-old running back, 48 (94%) of surgeons selected bone–patellar tendon–bone (BPTB) autograft, whereas 3 (6%) chose quadriceps tendon autograft. No surgeons selected hamstring autograft or any allograft. For their 25-year-old recreational athlete (question 4), 48 (94%) of surgeons recommended autograft, 2 selected allografts, and 1 said it was patient dependent. Of the surgeons who chose autograft, 33 (65%) chose BPTB autograft (Fig 2). For the 35-year-old recreational athlete, 20 (39%) chose a BPTB autograft, 13 (25%) chose a quadriceps tendon autograft, and 11 (22%) chose a hamstring autograft (Fig 3). In the setting in which a 20-year-old running back sustained an acute proximal ACL tear, 49 (96%) would choose ACL reconstruction, whereas 2 (4%) would choose a primary ACL repair.

In total, 71% of surgeons performed anteromedial portal (flexible or rigid reamer) for drilling the femoral tunnel, 8 (16%) performed retrograde drilling, and 7 (14%) performed transtibial technique (Fig 4). Question 7 asked if a lateral extra-articular tenodesis or anterolateral ligament surgery be performed in addition to their ACL surgery in a primary ACL reconstruction of a starting running back. In total, 43 (84%) said yes, they would, but only if certain findings are noted such as a Segond fracture, revision surgery, high-grade pivot. When asked if they recommend performing initial physical therapy with an outpatient physical therapist to supplement the team sports medicine staff, 38 (75%) recommended just working with the team staff only whereas 13 (25%) recommended supplementing with an outpatient physical therapist. When asked their criteria of permitting their starting running back to return to full competition, 15 (29%) of surgeons said return to sport strength and agility tests had to have been passed, whereas 13 (25%) said those same tests should be passed and a minimum of 9 months postoperatively had passed. Twelve (24%) formally recommend the use of a brace after ACL reconstruction whereas 38 (76%) did not.

Compared with the 2014 study by Erickson et al.,[4] which surveyed high-level American football physicians, several notable alterations in preferences were appreciated after a decade. Most notably is the decrease in preference for hamstring autograft and the increase in use of quadriceps tendon autograft. In 2014, 42.3% of surgeons preferred hamstring autograft for their 25-year-old athlete; however in 2024, this decreased to 8%. In the 35-year-old athlete, 45.3% of surgeons preferred hamstring autograft in 2014 compared with 22% in 2024 (Fig 5).

Patella tendon autograft remained the favorite for the 20-year-old running back (86.1% in 2014 and 94% in 2024), the 25-year-old recreational athlete (49.6% in 2014 and 65% in 2024), as well as the 35-year-old recreational athlete (39%).

In 2014,[4] 67% of surgeons performed anteromedial (AM) portal drilling techniques, 25% transtibial, and 8% used the 2-incision technique. Retrograde drilling was not reported as a preference at that time. In 2024, AM

Football Team Physician Survey Questionnaire

1. In your career, how many years have you performed ACL reconstructions?

2. In 2022, approximately how many ACL reconstructions did you perform?

3. During ACL reconstruction for a 20-year-old starting NCAA Division 1 or NFL running back, what is your preferred graft choice?
a. Bone-patellar-tendon bone (BPTB) autograft
b. BPTB allograft
c. 4-strand semitendinosus-gracilis autograft
d. Quadriceps tendon autograft
e. Achilles tendon allograft
f. Anterior tibial allograft
g. Other (please specify)

4. During ACL reconstruction for a 25-year-old recreational athlete, what is your preferred graft choice?
a. BPTB autograft
b. BPTB allograft
c. 4-strand semitendinosus-gracilis autograft
d. Quadriceps tendon autograft
e. Achilles tendon allograft
f. Anterior tibial allograft
g. Other (please specify)

5. During ACL reconstruction for a 35-year-old recreational athlete, what is your preferred graft choice?
a. BPTB autograft
b. BPTB allograft
c. 4-strand semitendinosus-gracilis autograft
d. Quadriceps tendon autograft
e. Achilles tendon allograft
f. Anterior tibial allograft
g. Other (please specify)

6. For an ACL tear involving a 20-year-old starting NCAA Division 1 or NFL running back who had an acute proximal ACL tear, would you recommend primary ACL repair or ACL reconstruction?
a. Primary ACL repair
b. ACL reconstruction

7. During ACL reconstruction for your starting running back, how do you most commonly perform femoral tunnel drilling?
a. Anteromedial portal
b. Transtibial technique
c. Retrograde reaming

8. During primary ACL reconstruction for your starting running back, would you recommend a lateral extra-articular tenodesis (LET) or anterolateral ligament (ALL) be performed in addition to the ACL regardless of ACL graft source?
a. Yes, definitely
b. No, never
c. Yes, but only if certain findings noted (Segond fracture, revision surgery, high-grade pivot, etc.)

9. For your NCAA or NFL starting running back undergoing ACLR, do you recommend they perform initial physical therapy with an outpatient physical therapist in addition to the team sports medicine staff or team staff alone?
a. Yes, outpatient PT in addition to team staff
b. No, team staff only

10. After ACL reconstruction in your starting running back, what criteria do you use to permit return to sports in a regular season competitive game-type setting? Please select all answers that apply.
a. After a minimum of 6 months postoperatively
b. After a minimum of 9 months postoperatively
c. After a minimum of 12 months postoperatively
d. After normal range of motion, no pain, good strength, and subjective stability are present
e. After a set of "return-to-sport" and strength tests has been completed and passed (e.g., Quad index >90%, hop testing shows good mechanics, etc)
f. Other (please specify)

11. Once your starting running back has returned to sport after ACL reconstruction, do you recommend use of a knee brace during sport?
a. Yes
b. No

Fig 1. Questionnaire sent to team knee surgeons of the National Football League.

portal techniques accounted for approximately 71%, transtibial 14%, and retrograde drilling 16% (Fig 6). No surgeons reported preferring 2-incision technique. There was a decreasing trend of recommending a derotational "sports brace" since 2014[4] for return to sport (36% in 2014 compared with 24% in 2024).

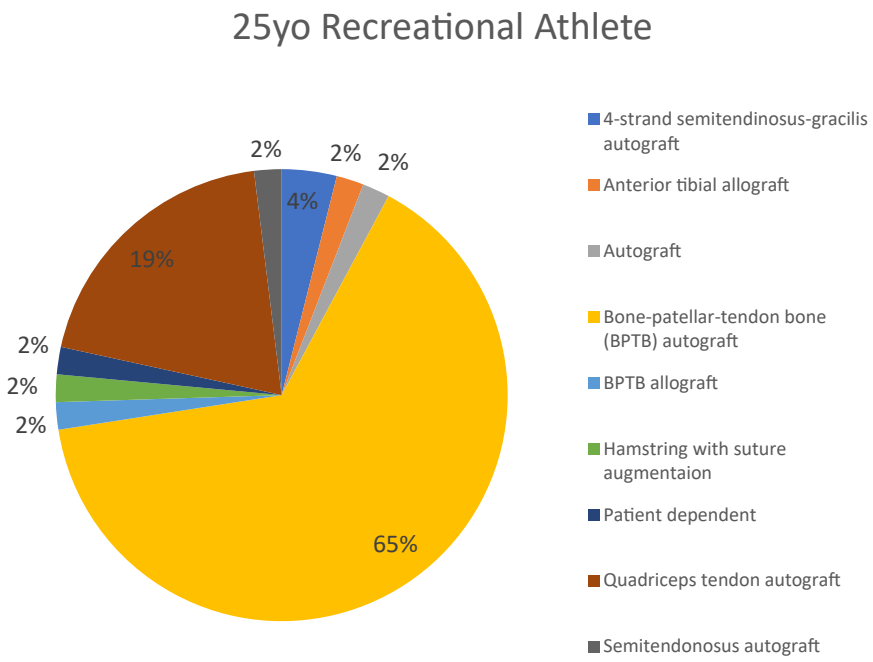

Fig 2. Surgeons' anterior cruciate ligament reconstruction graft preferences for a 25-year-old recreational athlete.

Discussion

In this study, we found that patella tendon autograft is the favored graft in ACL reconstruction, with quadriceps tendon being second most favored. Lateral augmentation procedures would be considered in a primary ACL reconstruction but only if certain findings are present. ACL repair would not be recommended in a higher-level athlete with a proximal ACL tear.

The increased preference of quadriceps tendon autograft (Fig 5) by NFL team knee surgeons does align with

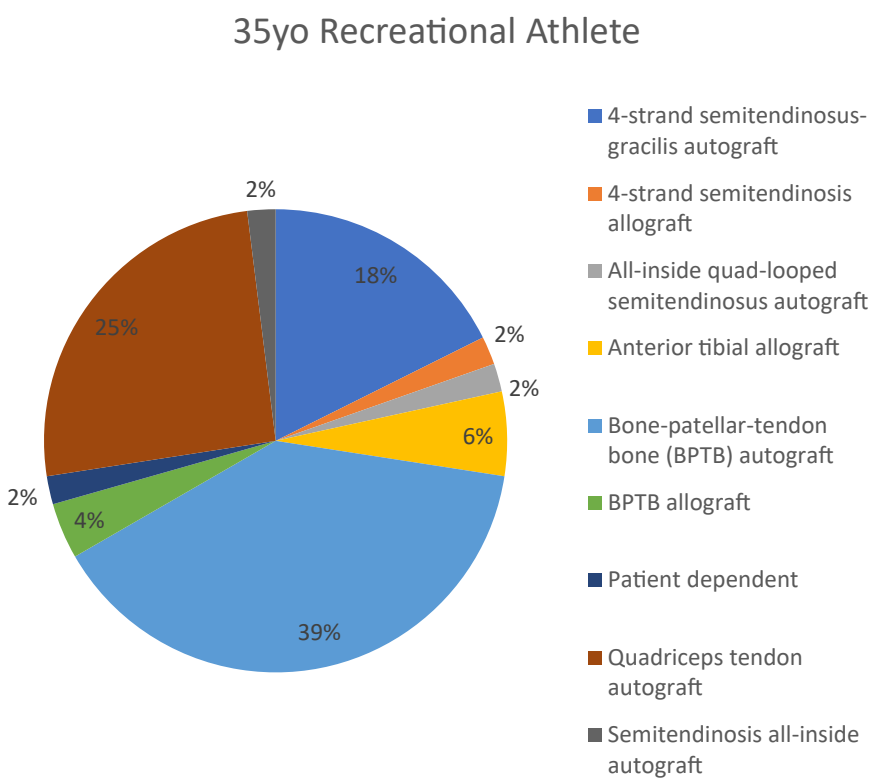

Fig 3. Surgeons' anterior cruciate ligament reconstruction graft preferences for a 35-year-old recreational athlete.

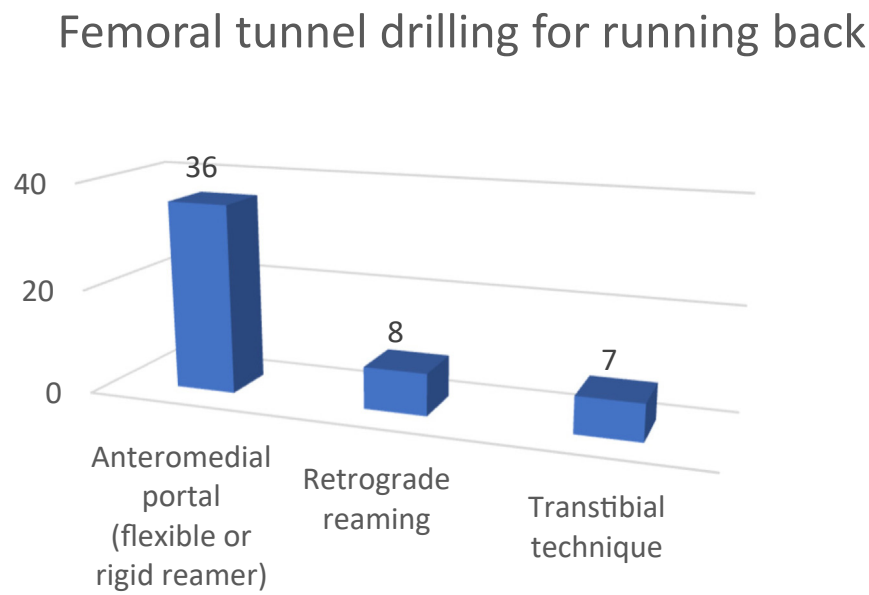

Fig 4. Surgeons' anterior cruciate ligament reconstruction femoral tunnel drilling technique preferences.

general trends by providers who are using quadriceps tendon autograft more commonly over the 2010s and 2020s.[5,6] Similarly, there seems to be more public interest in quadriceps tendon autografts as well, with one study showing an Internet search engine investigation from 2008 to 2019 that had quadriceps tendon autograft showing increasing searches more than patella tendon or hamstring autografts.[7] Despite this, patella tendon autograft remained the most preferred graft for the 20-year-old running back (86.1% in 2014 and 94% in 2024), the 25-year-old recreational athlete (49.6% in 2014 and 65% in 2024), as well as the 35-year-old recreational athlete (39%), as noted previously.

Multiple studies investigated the biomechanical advantages of various techniques drifting away from transtibial techniques to others, although there were learning curves initially with transitioning.[8-10] When the studies were published, there was interest in the

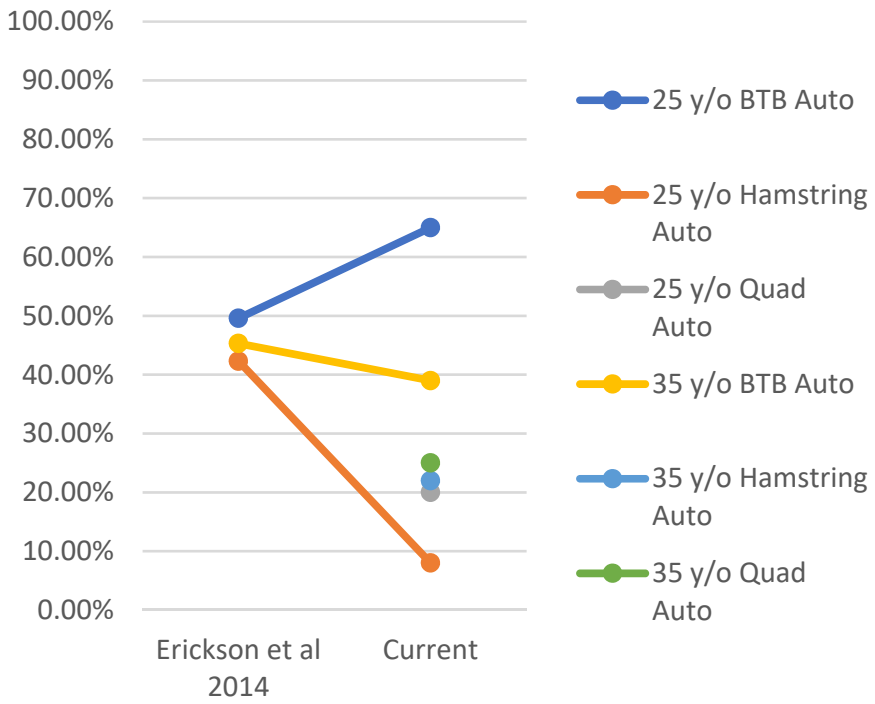

Fig 5. Comparison of graft preference of surgeons of National Football League by athlete's age, with trends shown from Erickson et al. 2014 to current investigation.

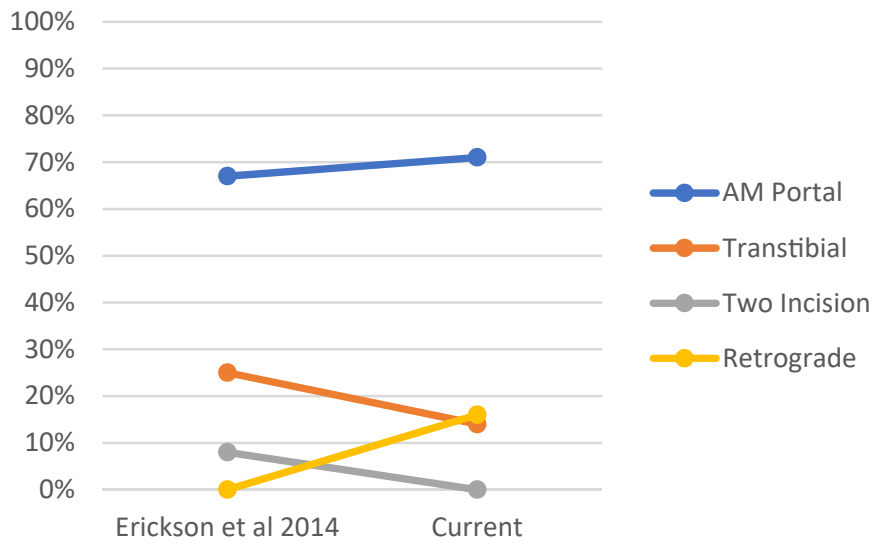

Fig 6. Trends of femoral tunnel drilling preferences by knee surgeons of National Football League.

2014 study[4] to see if those various techniques would be performed by NFL and NCAA surgeons who may see these ACLs tested at varying degrees. In this investigation, we sought to see whether there any further development in drilling preferences over a decade. What we saw in this investigation was that AM portal techniques have largely stayed similar from 2014[4] to 2014 (67% to 71%), but transtibial has decreased (25% to 14%) and retrograde reaming has increased with 16% of NFL knee surgeons using this technique.

As more research and literature has emerged, including the use of more modern techniques, there has been a renewed interest and implementation of primary repair for proximal ACL tears.[11,12] Although the results of primary ACL repair continue to emerge, there is still a question of appropriate indications largely by age or activity level, with the authors of recent meta-analysis showing a 6-fold increase in revision ACL surgery compared with ACL reconstruction in patients ages 21 and younger.[13] In this investigation, when knee surgeons in the NFL were presented with the idea of a primary ACL repair in their 20-year-old running back, 49 of 51 (96%) said they would not recommend this.

Lateral extra-articular augmentation procedures (LEAP) for instability prevention have been described in the past but largely had been abandoned for an extended period.[14] There was increased interest with the further description of the anterolateral ligament into 2013[15] and subsequently with the results of the STABILITY trial beginning in 2020.[16,17] In the early 2020s, this has been one of the larger trends and discussion points for various types of clinical situations, namely greater risk athletes and particular clinical situations such as revision ACL surgery, patients with significant hyperextension, and others. In one 2024 published investigation asking surgeons in 2016, 2018, 2020, and 2023 about their use of LEAP procedures, respondents increased their use approximately 10% over a 7-year period.[18] In a 2024 survey of head team NFL physicians, 6 of 31 (19%) said they would consider a LEAP in a primary ACLR, although in the body of the manuscript it stated that "none of the respondents regularly used an anterolateral ligament or lateral-extra articular tenodesis in the primary setting."[19]

The Delaware-Oslo cohort paper in 2016 was a seminal paper for ACL research, citing sooner than 9 months as a significant influencer of risk for reinjury as well as ability to pass functional testing including strength and agility.[20] When asked about what benchmarks should be met before returning to competitive play, 15 (29%) of surgeons said return to sport strength and agility tests had been passed, whereas 13 (25%) said those same tests should be passed and a minimum of 9 months post-operatively.

Use of a derotational/functional stabilizing brace has been used in some form since the 1960s, but there has been limited evidence on its effectiveness on injury prevention for ACL injuries.[21-24] In a 2000 evaluation in which 12 healthy knees were evaluated with a brace then placed through a battery of physical tests, there was a significant difference in single-leg and timed hop distances, with the braced group having decreased distance of a single-leg hop and increased time to perform a timed hop.[22] In a study from 2020, 28 patients who underwent ACL reconstruction completed various physical tests when they were released to sport and then 3 months later with and without a brace. The authors of the study concluded that wearing a brace did not hinder physical performance.[4] In our current evaluation of what NFL team knee surgeons recommend, 24% recommended a functional brace upon returning to sport. In 2014, this was 36%.[4]

This type of survey provides a general overview and does not capture the complex decision-making that can occur in clinical practice. An example of this is the question of supplementing their ACL surgery with a LEAP in the primary setting. Although >80% of surgeons said they would consider this, it is only if there were secondary characteristics present and thus is not a blanket statement of suggestion in a primary ACL surgery.

One strength of this study is that we limited the respondents to practicing sports medicine knee surgeons rather than a "team physician" who does not perform ACL surgery or see ACL injuries in their normal practice. In addition, we tried to model the survey like the 2014 study but amending it to include hot topics in sports medicine knee surgery such as ACL repair and lateral augmentation procedures.

The respondents of this study included 51 NFL team knee surgeons. When reviewing other polls of NFL team physicians, these have included 32 or fewer team physicians.[19,25,26] Although our 53% respondent rate may be considered less than optimal, our subspecialized total physician respondents is still greater than these other studies. This 53% number is also greater than the 51% average respondent rate by healthcare professionals noted in a 2020 metanalysis of respondence rate to surveys.[27]

Limitations

This study is not without limitations. The primary limitation of this study is that NCAA physicians were not included in this poll. Second, the total number of respondents is not as significant, limiting the ability to compare our study with the 2014 study by Erickson et al.[4] The response rate in our study is low, which creates a high potential for responder bias. Further limitations of this study are lack of advanced statistical analysis and the fact that the survey itself is not validated. Like any study of a select group of providers, the results of this survey is not necessarily generalizable.

Conclusions

Patella tendon autograft is the favored graft in ACL reconstruction among professional and recreational athletes by NFL team orthopaedic knee surgeons. Quadriceps tendon autograft is the second most preferred graft. Lateral augmentation procedures would be considered in a primary ACL reconstruction but only if certain findings are present. Most respondents do not recommend supplementing rehab with outpatient PT or using a brace for return to play.

Disclosures

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: P.A. was a participant in the involved survey. All other authors (S.H., B.B., M.L., D.L.) declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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